

Coin Appraisal Register

Build Specification — M1 + M2 + M3 Protection Layer

Authoritative scope for the engagement. The work to be quoted is defined by Sections 1 through 8 below. M1 acceptance is a precondition to M2 + M3 kickoff payment.

Engagement components

- **M1 — Three fixes** — \$175 already contracted, held pending acceptance test. Section 1.
- **M3 PWA Infrastructure** — \$350 already contracted, delivered, live. No change.
- **M2 — Full scope** — Section 2. To be quoted.
- **M3 — Protection layer** — Section 3. To be quoted.

1. M1 — Deliverables and Acceptance

Deliverables

- Bulk calculator extensions covering the coin types contracted under M1.
- Numismatic grading selector: Circulated / Uncirculated / Slabbed with per-condition multipliers from Config_CoinTypes.
- Custom numeric keypad on iPad: decimal, backspace, clear, tap-outside to dismiss.
- Running totals: Total Value and Total Offer, displayed at all times.

Acceptance criteria

M1 passes when, on the deployed staging URL tested on iPad Safari, each coin type within M1's contracted scope calculates correctly; weight-based items calculate correctly across a range of gram weights including decimals; the grading selector applies the correct multiplier; the keypad behaves correctly; and running totals update in real time and zero out when all items are removed.

\$175 released same-day on acceptance. M2 + M3 kickoff payment is not wired until M1 has passed. Please confirm a target date.

2. M2 Scope

All features below are in scope for M2. Please price each subsection individually in the revised SOW.

2.1 Customer Intake

Feature	Description
Intake form	Five required fields: name, phone, DOB, zip, TCPA opt-in. All must be completed before the calculator opens.
Single lot photo at intake	One photo of the customer's entire haul (NOT per-item). Uploaded to cloud storage; URL written to Collection_Photo_URL.
Driver's license capture	Photo upload (uploaded to cloud storage, URL stored on deal record) plus manual entry of DL number. No PDF417 OCR — manual entry is the default.
Under-18 hard stop	DOB-based. Deal cannot proceed. Hard block, no override.
Standalone zip field	Zip is independently queryable for geographic analytics without exposing the full address.
Returning customer lookup	On intake, app searches Customer_Master by phone + DOB. If matched, prior history is loaded and visible to the rep. New record created otherwise.

2.2 Item Entry and Pricing Engine

Feature	Description
All M1 coin types	Bulk calculator for the M1 types. Already partially live.
Bullion coins (8 types)	Unit × spot + configurable premium per type. Premiums from Config_CoinTypes.
Sterling silver flatware	Gram × purity × spot. Hallmark popup blocks weight entry until acknowledged.
Foreign silver coins (4 entries)	Stored purity factors + manual override selector for the fourth entry.
Paper money	Manual entry: denomination, year, condition, offer. Static manager-review note always visible.
Numismatic grading selector	Circulated / Uncirculated / Slabbed, per-condition multipliers from Config_CoinTypes. Extends M1 logic to all graded categories.
Pre-1933 US Gold red hard stop	Full-screen stop with timestamped manager-PIN acknowledgment written to the deal record (see Section 3.2). Cannot be skipped or back-buttoned.

2.3 Lookup Integrations

Feature	Description
PCGS CoinFacts	Cert authentication via Make.com HTTP. PCGS API key live before M2 starts.
CDN Greysheet Dealer+	Wholesale pricing via Make.com HTTP. CDN paid subscription with API tier live before M2 starts.
Fallback chain	CDN → Manual_Price_Override → Ask Manager. Side-by-side display of CDN price and offer at configured margin.
Manual cert entry	Cert number entered manually; PCGS API validates after entry. No barcode-scanner SDK in scope.
eBay sold-listings comps via Apify	Low / median / high range logged on every lookup. Empty results valid. Apify paid plan, 50 runs/day minimum.
Metals spot price feed	Live gold and silver via Metals-API. Platinum included.

2.4 Dual Pricing Engine

Extends the M1 calculator. Each item carries a Max Payout (calculated ceiling) and an Actual Offer (rep-entered). Negotiation Delta = Max Payout – Actual Offer. Roll-up totals at deal level.

Feature	Description
Max Payout per item	Per-item ceiling derived from the item's underlying calculated value (per Section 2.2 or lookup integrations per Section 2.3) minus the margin in Config_Margins for that category.
Actual Offer per item	Rep-entered. Must be $\leq$ Max Payout unless an override is applied (Section 2.13).
Negotiation Delta	Max Payout – Actual Offer. Visible to rep only, never on customer view.
Roll-up totals	Total Max Payout, Total Actual Offer, Total Delta. Extends M1 running totals.
Margin % display	Live margin percentage from Total Actual Offer relative to Total Max Payout. Rep view only.

2.5 Deal Summary and Customer View

Feature	Description
Internal deal summary	Rep-facing: line items, totals, margin %, Accept / Decline. Margin visible to rep only.
Swipe to customer summary	Swipe gesture transitions to a clean customer-facing view that hides margin information. Same screen flow; no separate customer UI.

2.6 Acceptance Flow

Feature	Description
Payment selector	Cash / check / wire / other. Written to deal record.
\$9,500 cash hard stop	Single cash transaction over \$9,500 triggers a full-screen stop. Manager PIN required to proceed; event logged.
Offer letter PDF	Generated client-side at acceptance. Includes deal terms, offer amount, customer info, signature line. Uploaded to cloud storage; URL written to deal record.
Customer signature capture	Touch-screen signature on iPad. Stored as PNG in cloud storage. Embedded in the offer letter PDF.
Lot photo at acceptance	Final photo confirming items received. Uploaded to cloud storage; URL written to deal record.

Feature	Description
Review prompt	App fires the GHL accepted-deal webhook with opt-in flag; review-request flow is handled inside GHL.

2.7 Decline Flow

Feature	Description
24-hour price lock	On decline, captures phone (if not on file) and creates a deal record with price_lock_24hr flag. Customer can return within 24 hours and the offer stands.
Declined webhook	Outbound webhook with deal_declined tag and price_lock_24hr where applicable.

2.8 Manager-Required Workflows

Flag system has two states: Normal (proceed) and Manager Required (deal pauses, manager PIN unlocks). The five Manager Required triggers are listed in Section 3.3.

Feature	Description
Ask Manager button	Permanent on every screen. Freezes the deal, notifies manager, writes timestamped note. Manager resumes via PIN.
Not Sure What This Is button	Sends item photo to manager, opens search of Config_CoinTypes reference content, saves the deal to a pending state until manager input is received.

2.9 GHL Integration

Feature	Description
Accepted-deal webhook	Outbound only. Full payload: customer info, TCPA opt-in flag, deal value, item tags, deal ID, payment type, photo URLs, signed offer letter URL.
Declined-deal webhook	Outbound only. Contact info, deal_declined tag, price_lock_24hr where applicable.
Direction	Outbound only. No bidirectional ID capture in scope.
Inside-GHL configuration	Not in scope. Workflows, SMS templates, follow-up sequences, and pipeline configuration are handled by the operator's GHL administrator.

2.10 Webhook Fan-Out

Every accepted or declined deal fans out independently to multiple destinations. Each webhook fires with independent retry logic; a failure in one does not block the others.

Destination	Purpose
Primary database	System of record. Receives the full deal payload.
GHL	Section 2.9 webhooks.
Zoho	Operational reporting destination. Receives the deal payload mapped to the operator's Zoho schema.
Configurable additional destination	Fourth webhook URL stored in Config_Operator. Fired with the same payload as the others.

2.11 Multi-Location and Roadshow

Feature	Description
Config_Locations	Per-deployment location records (name, default branding placeholders, default config). Admin-editable.
Location-scoped deals	Every deal carries a Location_ID populated from the active Config_Locations record at deal start.
Shared rep pool	Reps exist once in Reps_Master and can be active at any location. The app tracks which location each deal happened at via Location_ID.
Config_Events	Roadshow event records: event name, start date, end date, associated location. Admin-editable.
Event-tagged deals	When the app is in event mode, every deal carries an Event_ID in addition to Location_ID. Null outside event mode.

Feature	Description
Event mode toggle	Manager-PIN-gated toggle that switches the app into event mode for the duration of the event.

2.12 Cloud Storage

All photos (intake lot photo, DL photo, acceptance lot photo, item photos) and all generated PDFs (signed offer letter) are stored in S3 or Cloudflare R2. URLs are written to the deal record. Platform-native attachment fields are not used.

2.13 Override Code Mechanism

Feature	Description
Single-use override code	6-digit code issued by manager. Bound to a specific deal. Expires 30 minutes from issuance. Single-use.
Use case	When a rep needs to enter an Actual Offer above the Max Payout, manager issues the code. Rep enters it; override is recorded on the deal record with the issuing manager.
Audit	Each override logged: deal ID, rep, issuing manager, item, amount over max, reason, timestamp.

2.14 Data Model

Feature	Description
Deal_Log table	All deal-level fields (35+) per the kickoff field map. Built before any Make scenario is wired.
Three attribution fields per deal	Rep_PIN_ID (who did the deal), Credited_Salesman (who it's credited to for commission), Referrer (any referrer for this deal).
Auto-populated Operator_ID and Location_ID	Driven from single-row config tables.
Spot price snapshot on deal completion	Gold and silver, captured at the moment of acceptance, written to the deal record.
Deal status workflow	States: Pending, Approved, Closed, Reconciled, Disputed.
Customer_Master table	Separate from Deal_Log. Keyed on phone + DOB. Stores returning-customer profile data.
Offline degradation	Cached spot prices used when the metals feed is unreachable. Webhooks queue and retry when signal returns.
Schema discipline	Use standard data types (string, integer, decimal, boolean, datetime). No business logic in database formulas; calculations run in app code. No reliance on platform-native automations — all workflow runs in Make.com scenarios.

2.15 Configuration Tables

Table	Purpose
Config_CoinTypes	Multipliers, premiums, purity factors, key date warnings, image URLs (if used).
Config_Margins	Margin % per category. Feeds Max Payout (Section 2.4).
Config_Operator	Single-row operator config: name, branding placeholders, default location, fourth webhook URL.
Config_Locations	Per-deployment location records (Section 2.11).
Config_Events	Roadshow events (Section 2.11).
Reps_Master	All reps with names and 4-digit PINs. Manager-tier flag.

2.16 Loom Walkthroughs

Delivery includes Loom walkthroughs covering every admin-editable Make.com setting and every configuration table, so a non-technical admin can update margins, coin types, locations, recipients, and rep records without breaking the build.

3. M3 Protection Layer

Sits on top of the PWA infrastructure already contracted at \$350 (delivered, live). All features below are in scope.

3.1 Authentication

Feature	Description
Rep PIN authentication	4-digit PIN, encrypted at rest in Reps_Master. 3-strike lockout for 5 minutes; manager alert on lockout.
Manager PIN tier	Separate manager PIN. Manager-tier flag in Reps_Master. Manager-only views for override issuance, approval gates, activity log, dashboard, and test mode toggle.
Auto-lock	App auto-locks after 2 minutes of inactivity. Re-auth via Face ID or PIN.

3.2 Two-State Flag System

Normal (proceed) and Manager Required (deal pauses, manager PIN unlocks). No intermediate tier.

Manager Required trigger	Description
Pre-1933 US Gold	Full-screen red hard stop (Section 2.2); manager PIN to acknowledge and proceed.
\$9,500 cash hard stop	Section 2.6. Manager PIN to proceed.
Actual Offer above Max Payout	Override code required (Section 2.13).
Ask Manager pressed	Section 2.8.
Not Sure What This Is pressed	Section 2.8.

3.3 Approval Gate

Feature	Description
Pending Approval screen	When a deal triggers Manager Required, rep sees a Pending Approval state until the manager acts.
Manager response mechanism	Short-interval polling (5-second interval) on the rep's Pending Approval screen. Manager acts via the manager dashboard.
Timeout behavior	If no manager action in 15 minutes, the deal returns to the rep with a manager-unavailable state. Resumable when a manager is available.

3.4 Manager Dashboard

Feature	Description
Per-rep performance	Deal count, total gross profit, average margin captured, average negotiation delta, override count — per rep, current quarter.
Override audit log	Every override visible: rep, issuing manager, item, amount over max, reason, timestamp.
Pending Approval queue	Live queue of deals awaiting manager action.
Override code issuance	Manager issues a single-use override code bound to a specific deal (Section 2.13).

3.5 Test Mode

Feature	Description
Manager-PIN-gated toggle	Manager switches the app into Test Mode. Deals created in Test Mode are written to Test_Deal_Log (not Deal_Log) and do not fire webhooks.
Visual indicator	Persistent banner across the top of the app when in Test Mode.

3.6 Picker

Feature	Description
Names only	Picker shows item names. No coin images, expected gram weights, or recently-used row.
Search and filter bar	Quick filtering by name or category.
Key date warnings	Pulled inline from Config_CoinTypes.key_date_warning.

3.7 Plain English Tooltips

Tap-to-reveal one-sentence explanations on Face Value, Melt Value, Times Face, Max Payout, Actual Offer, and Negotiation Delta. Plain English, written for non-numismatist customers viewing the screen.

4. Signature Implementation

Feature	Description
Touch-screen capture	Customer signs on the iPad. Canvas accepts touch and stylus.
Stored as PNG	Signature saved as PNG, uploaded to cloud storage. URL on deal record (Signature_Image_URL).
Embedded in offer letter PDF	Acceptance PDF (Section 2.6) embeds the signature inline above the signature line.
Audit	Capture event timestamped on the deal record.

5. Boundary of Engagement

Scope is defined by Sections 1 through 8. Anything not specified in those sections is not in scope. The following are handled by other parties or otherwise out of scope:

- **Inside-GHL configuration.** Workflows, SMS templates, follow-up sequences, opportunity pipeline configuration.
- **CRM-side configuration.** Mapping, dashboards, and reports in the operator's primary CRM.
- **Hardware procurement.** iPads, stands, scales.
- **Branding finalization.** Logo, accent color, app name. Placeholders during build; final assets land via follow-up update.
- **Recurring third-party costs.** PCGS, CDN Greysheet, Metals-API, Apify, Make.com Pro, Vercel, primary database, cloud storage. Operator costs.
- **Compliance automation.** State law-enforcement reporting, structuring detection, Form 8300 automation, 1099-B threshold tracking and filing, encrypted PII vault. Not in scope.
- **Native iOS push notifications.** Not in scope. 5-second polling (Section 3.3) is the manager-notification mechanism.
- **Post-launch new features.** The M3 PWA milestone includes 3 months of bug-fix support. New features are separate scope.

6. Pre-Kickoff Requirements

The following must be confirmed before M2 + M3 kickoff payment is wired:

- M1 passed acceptance per Section 1. \$175 released.
- Revised SOW signed with itemized pricing per subsection of Sections 2 and 3.
- PCGS CoinFacts, CDN Greysheet Dealer+, Metals-API, and Apify (paid plan, 50 runs/day minimum) returning data in Make.com under operator credentials.
- S3 or Cloudflare R2 account provisioned with a dedicated bucket; access keys available.
- Photo + manual DL number entry confirmed as the capture path. No paid OCR SDK in scope.

Configuration data provided by operator at kickoff:

- Coin type configuration spreadsheet (names, categories, calculation methods, multipliers, premiums, purity factors, key date warnings).
- Config_Margins values per category.
- Config_Operator values (operator name, branding placeholders, default location, fourth webhook URL).
- Config_Locations values per deployed location.
- Config_Events values for known roadshow events.
- Reps_Master entries (names, 4-digit PINs, manager-tier flags).
- GHIL webhook URLs (accepted and declined endpoints) and Zoho webhook URL.
- Field map for the deal payload across destinations.

7. Payment Structure

Four milestones across M2 and M3 protection layer. Percentages apply to the combined revised SOW total. M1 (\$175) and M3 PWA infrastructure (\$350) are separate and already contracted.

Milestone	Trigger / %
1. Kickoff	API access confirmed in writing, cloud storage account live, SOW signed, M1 accepted. 20% .
2. Deal Flow + Dual Pricing Demo	M2 deal flow demo with dual pricing (Max Payout, Actual Offer, Negotiation Delta) working end-to-end on iPad. 30% .
3. M2 iPad Acceptance	Full M2 scope tested on iPad in production environment. 25% .
4. M3 iPad Acceptance	Full M3 protection layer tested on iPad. Loom walkthroughs delivered. 25% .

8. Requested in Revised SOW

- Target date for M1 acceptance.
- Itemized pricing per subsection of Sections 2 and 3.
- Confirmation of the four-milestone payment structure (Section 7).
- Realistic timeline per phase.
- Confirmation that hand-off boundaries in Section 5 are understood.

End of specification.